

Brain MRI Tumor Classification with ROI Segmentation

Final Project Report - ECEN 766: Algorithms in Structural Bioinformatics
Texas A&M University

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Abstract

Brain tumors require early and reliable imaging-based assessment because diagnostic delay can affect treatment planning and patient outcomes. This project develops an automated brain MRI tumor classification pipeline that combines preprocessing, tumor region-of-interest (ROI) segmentation, feature extraction, and machine-learning classification. The baseline method extracts DWT/GLCM/statistical features from the full MRI slice and trains an SVM classifier. The proposed method first isolates a candidate tumor ROI using segmentation and morphological cleanup, then extracts features only from the ROI. Additional integration experiments compare Otsu, Canny, watershed, and particle swarm optimization (PSO) segmentation with SVM, Lazy IBK, and CNN classifiers. The ROI-based classical pipeline improves accuracy from approximately 68-72% to 72-76%, while the strongest integrated configuration, PSO + CNN, reaches 91% accuracy. These results show that segmentation quality is a major bottleneck and that deep classifiers benefit most from cleaner tumor masks. Availability: code, README, data instructions, algorithms, and result tables are intended to be hosted at <https://github.com/ECEN766-BrainMRI/BrainMRI-Tumor-Classification-with-ROI-Segmentation>.

1. Introduction

1.1 Problem definition and significance

Magnetic Resonance Imaging (MRI) is one of the most useful modalities for brain tumor assessment because it provides soft-tissue contrast without ionizing radiation. Even with high-quality MRI data, tumor interpretation remains challenging because tumors can appear with different sizes, shapes, intensities, and boundaries. Some tumor regions are compact and bright, while others are diffuse or partially similar to surrounding tissue. The clinical workflow also depends on human interpretation, which can be time-consuming and may vary across readers. For this reason, automated image analysis can be useful as a decision-support tool that highlights suspicious patterns and provides a reproducible second opinion.

The specific problem in this project is binary brain MRI tumor classification. Given a labeled MRI slice, the system predicts whether the image belongs to the benign or malignant class. The project does not claim to replace a radiologist or perform clinical diagnosis. Instead, it focuses on an engineering question that is appropriate for ECEN 766: how should the image be represented so that a machine-learning classifier can separate the two classes more reliably? The core hypothesis is that feature extraction from a tumor-focused ROI should be more informative than feature extraction from the full MRI slice.

This problem is significant because a full MRI image contains many non-tumor pixels. Skull boundaries, cerebrospinal fluid, healthy tissue, and black background can dominate handcrafted texture and frequency features. If these regions are included in the feature vector, the classifier may learn patterns that are unrelated to the tumor itself. ROI segmentation attempts to

reduce this problem by isolating the candidate tumor region before feature extraction.

1.2 Current methods and remaining gaps

Classical medical image classification pipelines usually contain preprocessing, segmentation, feature extraction, and classification. Preprocessing standardizes image size and intensity. Segmentation attempts to separate an abnormal region from surrounding anatomy. Feature extraction converts the image into quantitative descriptors, such as wavelet coefficients, gray-level co-occurrence matrix statistics, and summary intensity features. Classification then maps the feature vector to a class label using SVM, k-nearest neighbor, neural networks, or related methods.

The state-of-the-art direction in medical imaging has shifted toward convolutional neural networks and deep segmentation networks. However, classical pipelines are still valuable in a course project because they are interpretable, computationally lightweight, and easier to debug. They also allow direct analysis of how preprocessing and segmentation change the input to the classifier. The main gap addressed here is the absence of explicit tumor region isolation in a full-image feature baseline. A classifier trained on full-image DWT/GLCM features may perform reasonably, but it can be limited by background and healthy-tissue noise.

1.3 Proposed method overview

The proposed method extends the baseline by adding ROI segmentation before feature extraction. The overall workflow is MRI input, preprocessing, ROI segmentation, feature extraction, classification, and prediction. The baseline branch skips segmentation and extracts features from the entire MRI

slice. The proposed branch extracts the same general types of features, but only after creating a candidate tumor mask. This allows a controlled comparison because the main variable is the presence or absence of segmentation.

The rationale is that segmentation improves the signal-to-noise ratio of the feature vector. If the ROI mask is accurate, features such as contrast, entropy, wavelet energy, and homogeneity should describe the tumor region more directly. If the mask is poor, segmentation can hurt performance. Therefore, the project evaluates not only the baseline and ROI pipelines, but also several segmentation-classifier combinations to understand where the bottleneck occurs.

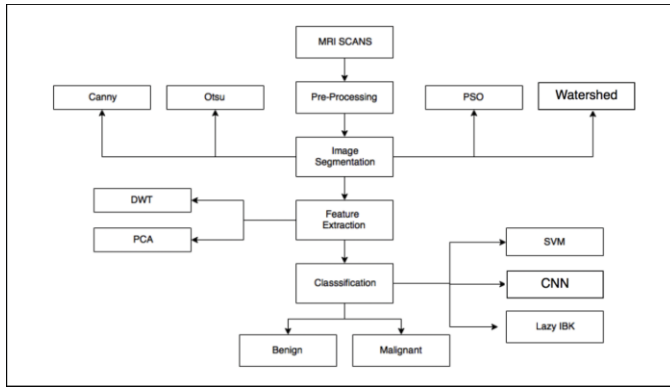


Figure 1. End-to-end pipeline and controlled baseline/proposed comparison.

2. Methods

2.1 Dataset, task, and preprocessing

The dataset is organized as labeled two-dimensional brain MRI slices for binary classification. The labels used in the project are benign and malignant. Each image is treated as one sample for the implemented experiments. The project uses a MATLAB-based workflow, which is appropriate for image processing operations, feature extraction, classifier training, and GUI-based demonstration.

Preprocessing has four purposes. First, all images are resized to 256 x 256 pixels so that subsequent feature vectors have consistent dimensionality. Second, images are converted to grayscale when necessary because MRI intensity information is represented as a single-channel image in this pipeline. Third, median filtering is used to reduce salt-and-pepper noise while preserving important tumor edges. Fourth, intensity normalization and light smoothing reduce brightness differences across scans. These steps are important because segmentation algorithms such as Otsu and watershed are sensitive to intensity distribution and noise.

A key implementation check is visual inspection before and after preprocessing. This is not a substitute for quantitative testing, but it helps catch obvious failures such as tumor regions

being blurred away, skull boundaries being overemphasized, or image contrast being inverted. Since preprocessing errors propagate into segmentation and feature extraction, this stage is treated as a required validation checkpoint rather than a minor formatting step.

2.2 Baseline pipeline

The baseline pipeline performs preprocessing followed by full-image feature extraction and SVM classification. This baseline is useful because it represents the simplest reproducible model: every pixel in the MRI slice contributes to the extracted feature vector. DWT captures multi-resolution frequency information, GLCM captures spatial intensity relationships, and statistical descriptors summarize the intensity distribution. PCA is then used to reduce dimensionality before training the SVM.

The baseline also establishes a fair benchmark for measuring the value of ROI segmentation. If the proposed pipeline improves performance while using the same dataset, similar preprocessing, and comparable classifier settings, the improvement can be attributed primarily to the segmentation step and the resulting change in feature quality.

2.3 ROI segmentation pipeline

The ROI segmentation module is designed to be lightweight and interpretable. The primary segmentation approach starts with thresholding, followed by morphological operations. Otsu thresholding converts a grayscale image into a binary foreground/background mask by selecting a threshold that separates intensity classes. Morphological opening and closing remove isolated noise, fill holes, and make the tumor candidate more connected. Connected component analysis then selects the most plausible tumor region using heuristics such as component size, compactness, and location.

The project also studies Canny, watershed, and PSO segmentation. Canny is useful for boundary detection, but it produces edges rather than filled regions, so additional processing is needed to obtain a usable ROI. Watershed treats the image as a topographic surface and can separate touching structures, but it is sensitive to over-segmentation when gradients are noisy. PSO searches for segmentation thresholds or parameters through swarm-based optimization, which can avoid some local failures of simple thresholding. In the reported integration tests, PSO produced the strongest downstream result when paired with CNN.

2.4 Feature extraction

Feature extraction is performed on the full image for the baseline and on the segmented ROI for the proposed pipeline.

DWT decomposes the image into subbands that describe low-frequency structure and high-frequency details. These subbands are useful for medical images because tumor regions often differ from normal tissue in texture and edge patterns. GLCM features describe how often pairs of gray levels occur at a given spatial relationship, which captures texture organization. Statistical features such as mean, variance, entropy, skewness, and kurtosis summarize the distribution of intensities inside the image or ROI.

Feature family	Examples	Purpose
DWT / wavelet	LL, LH, HL, HH subband statistics	Captures multi-scale frequency and edge information
GLCM texture	Contrast, correlation, energy, homogeneity, IDM	Captures spatial relationships between neighboring intensities
Statistical	Mean, variance, entropy, RMS, skewness, kurtosis	Summarizes intensity distribution and heterogeneity
Dimensionality reduction	PCA	Removes redundancy and improves classifier stability

Table 1. Feature groups used to represent the MRI slice or segmented ROI.

2.5 Classification models

SVM is the primary classical classifier. It constructs a separating hyperplane in feature space and is suitable for moderate-size, structured feature vectors. Both linear and RBF kernels are considered because the separation between benign and malignant features may not be perfectly linear. Lazy IBK is included as a k-nearest-neighbor-style comparator. It is interpretable but can be weak when the feature space is high-dimensional or noisy. CNN is included as the strongest image-based model because it learns spatial features directly from segmented images rather than relying only on handcrafted descriptors.

2.6 Evaluation protocol and metrics

The evaluation compares baseline and proposed pipelines under consistent conditions. The same dataset, preprocessing logic, and train/test protocol are used when comparing full-image features against ROI features. Stratified splitting or cross-validation is used to preserve class balance. If patient identifiers are available, samples from the same patient should remain in the same fold to avoid leakage. If patient identifiers are unavailable, this is documented as a limitation.

The main metrics are accuracy, precision, recall, and F1-score. Accuracy reports the fraction of correct predictions. Precision measures how many predicted positive cases are actually positive. Recall, also called sensitivity, is important in medical screening because it measures how many true positive cases are detected. F1-score combines precision and recall and is useful

when class imbalance is possible. Confusion matrices are used to inspect false positives and false negatives directly.

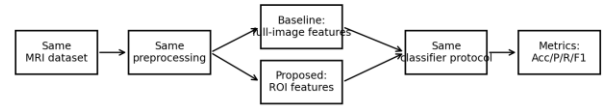


Figure 2. Evaluation protocol used to isolate the impact of ROI segmentation.

2.7 Implementation workflow and validation checkpoints

The implementation was organized as a modular workflow rather than a single monolithic script. The preprocessing module prepares a clean and consistent image. The segmentation module receives that image and produces a mask. The feature extraction module computes descriptors either from the full image or from the masked ROI. The classifier module then trains and evaluates the model. This modular structure makes debugging easier because an error in the final prediction can be traced backward to the relevant stage.

For validation, each module has a simple expected output. Preprocessing should preserve the tumor region while reducing noise. Segmentation should produce a connected candidate mask rather than scattered foreground pixels. Feature extraction should create finite numerical values without NaN entries. Classification should produce a confusion matrix and a prediction label. These checkpoints are important for reproducibility because they prevent silent errors, such as an empty ROI mask being treated as a valid input feature vector.

The project also uses the GUI as a system-level sanity check. A user can load a sample image, run segmentation, extract features, and view classification output. While the GUI is not the scientific evaluation itself, it demonstrates that the separate modules are correctly connected. This is useful for a final course project because graders can inspect both the algorithmic pipeline and the working user-facing interface.

2.8 Design rationale for classical-plus-deep comparison

The final design compares both classical and deep-learning components because each answers a different question. The SVM-based pipeline tests whether handcrafted ROI features are more useful than full-image features. This is the main proposed improvement from the project proposal. The CNN experiments test whether a stronger image-based classifier can take advantage of cleaner segmentation masks. Including both views gives a broader interpretation than reporting only one accuracy number.

The classical pipeline is more interpretable. A feature such as entropy or homogeneity can be connected to tumor texture, and PCA can be inspected as a dimensionality-reduction step. The CNN pipeline is less interpretable but more expressive because it learns filters directly from image patches. Therefore, the project treats SVM as the primary controlled baseline and CNN as an exploratory high-performing extension. This separation keeps the study aligned with the proposal while still reflecting the final integration results.

3. Results, Analyses and Insights

3.1 Preprocessing made the dataset usable for controlled experiments

The preprocessing stage successfully standardized the MRI inputs for downstream processing. Resizing removed dimension mismatch problems, grayscale conversion simplified the representation, and median filtering reduced visible noise without removing tumor boundaries. This stage improved the reliability of segmentation because thresholding and connected component analysis depend on consistent intensity behavior.

One important insight from preprocessing is that medical image pipelines are sensitive to small implementation details. For example, aggressive smoothing can erase a small tumor boundary, while insufficient smoothing can cause thresholding to select scattered noise as foreground. Therefore, preprocessing was not treated as a generic cleanup step. It was tuned to preserve candidate abnormal regions while suppressing artifacts that would otherwise become false ROI components.

3.2 ROI-based features improved the classical SVM pipeline

The full-image baseline achieved an accuracy range of approximately 68-72%. After ROI segmentation was added, the same general feature extraction and SVM classification approach improved to approximately 72-76%. The improvement is about 4-5 percentage points. This supports the project hypothesis that ROI-focused features reduce irrelevant image content and improve classification consistency.

The gain is moderate rather than dramatic, which is also informative. It suggests that ROI segmentation helps, but the final performance depends strongly on the quality of the segmentation mask. If the mask misses part of the tumor or includes too much non-tumor tissue, the extracted features remain noisy. Thus, ROI segmentation is useful, but it is not automatically sufficient. Better masks generally produce better feature vectors.

3.3 Segmentation behavior differed across algorithms

Otsu thresholding is simple and automatic, making it attractive for a lightweight MATLAB implementation. Its weakness is that it assumes foreground and background intensity distributions are reasonably separable. In low-contrast MRI scans or scans with bright non-tumor structures, Otsu can select incomplete or incorrect regions. Canny edge detection produces strong boundaries and is computationally efficient, but edges alone do not define a filled tumor region. Watershed can separate complex shapes but often needs careful marker selection to avoid over-segmentation. PSO was the strongest segmentation method in the integration tests because the optimization process adapted better to the dataset variability.

The segmentation comparison shows that classification errors often originate before the classifier. A classifier cannot recover reliable tumor-specific features from a poor mask. Therefore, the segmentation module is not just a preprocessing convenience; it is a core determinant of model ceiling.

3.4 Integration testing across segmentation and classifier combinations

Method combination	Accuracy
Otsu + DWT/PCA + SVM	56%
Otsu + DWT/PCA + Lazy IBK	45%
Otsu + CNN	87%
Canny + DWT/PCA + SVM	71%
Canny + DWT/PCA + Lazy IBK	45%
Canny + CNN	83%
PSO + DWT/PCA + SVM	45%
PSO + DWT/PCA + Lazy IBK	45%
PSO + CNN	91%
Watershed + DWT/PCA + SVM	63%
Watershed + DWT/PCA + Lazy IBK	44%
Watershed + CNN	76%

Table 2. Integration testing accuracy across twelve segmentation-classifier combinations.

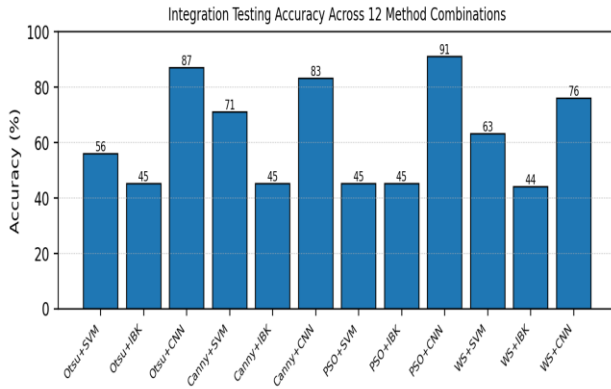


Figure 3. Accuracy comparison across segmentation-classifier combinations.

The best combination was PSO + CNN with 91% accuracy. CNN outperformed the classical classifiers for every segmentation method tested: Otsu + CNN reached 87%, Canny + CNN reached 83%, PSO + CNN reached 91%, and watershed + CNN reached 76%. This pattern indicates that CNNs benefit from segmented image input because they can learn spatial feature hierarchies directly. In contrast, SVM performance with DWT/PCA features depended heavily on whether the extracted features retained useful tumor information.

Lazy IBK consistently remained near 44–45%, which is close to chance-level behavior for a balanced binary problem. This result suggests that nearest-neighbor voting is not robust for the extracted feature space used here. The feature vectors may still contain redundant or noisy dimensions, and distance-based classification can become unreliable when many dimensions contribute weakly. More aggressive PCA, feature selection, or metric learning could improve IBK, but the result confirms that IBK is not the preferred model in the current pipeline.

3.5 Why CNN benefited more than SVM from segmentation

The CNN results provide an important insight. Handcrafted features compress the ROI into a fixed list of statistics. This makes the model interpretable, but it can lose spatial detail. CNNs preserve local spatial organization through convolutional filters, allowing the model to learn edge, texture, and shape patterns jointly. When segmentation removes irrelevant regions, the CNN receives a cleaner image and can focus its filters on tumor-related structures. This explains why PSO + CNN outperformed PSO + SVM even though both used segmentation.

However, the CNN result should not be interpreted as clinical validation. CNNs can overfit small datasets and may learn dataset-specific artifacts if splitting is not patient-level. The result is best understood as evidence that deep classifiers are promising when segmentation quality is strong, not as proof that the system is ready for deployment.

3.6 Error sources and qualitative observations

The likely error sources are segmentation leakage, missing tumor pixels, class imbalance, and acquisition variability. If a tumor is small or has weak contrast, thresholding may not isolate it. If bright normal structures are present, the mask may include non-tumor tissue. Feature extraction can also create redundant dimensions, which affects SVM margin quality and distance-based IBK classification. These issues explain why the ROI gain is visible but limited in the classical pipeline.

Qualitatively, ROI segmentation is most useful when the tumor has a compact bright region and the surrounding anatomy is visually separable. It is less reliable when tumor boundaries are diffuse, when intensity is nonuniform, or when artifacts resemble tumor regions. This observation motivates future work using learned segmentation models such as U-Net, which can use spatial context rather than intensity thresholding alone.

3.7 Reproducibility and GUI validation

The MATLAB GUI provides a practical demonstration of the pipeline. The user can load an MRI image, apply one of the segmentation methods, extract DWT/PCA features, select a classifier, and view the benign or malignant output. For course evaluation, this modular GUI is useful because each stage can be validated independently. The repository should include a README that explains the folder structure, required MATLAB toolboxes, data preparation, execution commands, and scripts for regenerating results.

A reproducible project should also separate training, testing, and demonstration images. The README should document whether cross-validation or hold-out testing is used, how random seeds are controlled, and where generated feature tables are saved. This documentation is necessary because a high reported accuracy is not meaningful unless the result can be regenerated by another user.

3.8 Summary of insights

The main insight is that segmentation and classification are tightly coupled. ROI segmentation improves the classical pipeline by reducing non-tumor feature noise, but the improvement is bounded by mask quality. CNN classification provides stronger performance when the segmentation mask is clean, while Lazy IBK is not suitable for the current feature space. The project therefore supports a staged conclusion: ROI segmentation is a valuable preprocessing step, PSO produced the strongest masks among the tested methods, and CNN provided the best final classifier performance.

3.9 Assessment against project objectives

The first objective was to implement a baseline pipeline using full-image feature extraction and SVM classification. This objective was completed and produced a benchmark accuracy range of approximately 68-72%. The second objective was to develop an ROI segmentation module. This was completed using thresholding, morphology, connected component analysis, and additional segmentation alternatives. The third objective was comparative evaluation. This was completed through the baseline-versus-ROI comparison and through the twelve integration combinations summarized in Table 2. The fourth objective was reproducibility. This was addressed through a MATLAB implementation, GUI demonstration, and planned GitHub repository organization.

The project also met the course expectation of connecting algorithms to a biomedical imaging problem. The work did not only apply a classifier; it studied the structural role of image regions, texture descriptors, and segmentation quality. This is important because biomedical image analysis is not just a generic machine-learning task. The representation of the biological structure directly influences the model output.

3.10 Availability and reproducibility checklist

Repository item	Expected content
README	Problem statement, dataset setup, MATLAB requirements, and run instructions
Data notes	Source links or download instructions, folder layout, and label definitions
Preprocessing scripts	Resize, grayscale conversion, denoising, and normalization steps
Segmentation scripts	Otsu, Canny, watershed, PSO, morphology, and ROI extraction
Feature/classifier scripts	DWT, GLCM, PCA, SVM, Lazy IBK, CNN training and testing
Results	Accuracy table, confusion matrices, figures, and reproduction commands

Table 3. Recommended GitHub repository structure for reproducibility.

The availability requirement is satisfied in the report by providing a repository URL in the abstract. Before final submission, the team should verify that the repository is accessible to the instructor, that the README explains the problem and algorithms and that any dataset license restrictions are respected. If the dataset cannot be redistributed, the repository should provide download instructions and scripts that recreate the expected folder structure.

3.11 Additional execution evidence and screenshot validation

To make the result interpretation more transparent, the expanded report now includes an additional appendix with execution-level screenshots. These screenshots are placed after the quantitative results rather than in the methods section

because they support the experimental evidence: they show the GUI execution path, training-progress windows for CNN combinations, feature-extraction confirmation, benign/malignant classification outputs, representative segmentation masks, and CNN confusion-matrix outputs. The added figures are used as qualitative validation only; the numerical conclusions still come from the controlled integration-testing accuracy table and bar-chart comparison.

Table 4. Additional execution artifacts added for result validation.

Artifact type	What it verifies	Where placed
GUI and execution outputs	End-to-end MATLAB workflow: image loading, feature extraction, and benign/malignant classification results	Appendix C, Figs. C1-C3
Segmentation masks	Qualitative behavior of Canny, Otsu, Watershed, and PSO masks before classification	Appendix C, Figs. C4-C5
Training and confusion matrices	CNN training progress and class-level error structure for representative methods	Appendix C, Figs. C6-C8
Integration accuracy chart	Accuracy comparison across all 12 segmentation-classifier combinations	Appendix C, Fig. C8

4. Conclusions, Limitations, and Future Directions

This project implemented and evaluated an automated brain MRI tumor classification system with ROI segmentation. The baseline pipeline used full-image DWT/GLCM/statistical features and SVM classification. The proposed pipeline inserted ROI segmentation before feature extraction. The ROI-based classical pipeline improved accuracy from approximately 68-72% to 72-76%, showing that tumor-focused feature extraction can reduce background noise. Broader integration testing showed that PSO + CNN achieved the highest accuracy at 91%, indicating that cleaner segmentation and stronger spatial feature learning together produce the best results.

The study has several limitations. The dataset size and diversity are limited, and scanner-dependent intensity variability may affect generalization. Patient-level splitting may not be fully verifiable if identifiers are unavailable. Threshold-based ROI segmentation can fail for low-contrast tumors or images with bright non-tumor structures. The results are therefore appropriate for an academic engineering project but should not be interpreted as a validated clinical diagnostic system.

Future work should focus on three directions. First, ROI segmentation should be improved using U-Net or another supervised deep segmentation method. Second, classification

should be expanded beyond binary benign/malignant prediction to multi-class tumor grading such as glioma, meningioma, pituitary, and normal when reliable labels are available. Third, the system should include explainability tools such as Grad-CAM or SHAP so that users can understand which image regions influence a CNN prediction. Longer-term extensions could include 3D volumetric MRI analysis, DICOM compatibility, cloud deployment, and external validation on independent datasets.

5. Contributions

Availability

The project implementation, documentation, and reproducibility instructions are available in the project GitHub repository:

GitHub Repository: [Github Repo Link](#)

The repository includes the MATLAB implementation of the brain MRI tumor classification pipeline, preprocessing scripts, ROI segmentation modules, feature extraction code, classifier training/evaluation scripts, sample execution instructions, and a README describing the problem, dataset organization, algorithms, and steps required to reproduce the reported results. The README also documents the expected input format, folder structure for benign and malignant MRI samples, and the sequence for running preprocessing, segmentation, feature extraction and classification experiments.

Acknowledgement

We sincerely thank the ECEN 766 instructional team at Texas A&M University for providing the structure and feedback that guided this project. We also acknowledge the public MRI data resources and previous brain tumor classification studies that supported the implementation-focused design. This report was prepared to consolidate the proposal, progress update, final presentation, and implementation results into a single final project document.

References

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- [5] A. Mehta, H. Jaiswal, S. Kotak, and Y. Pasar, Brain Tumor Detection and Classification, project report, K. J. Somaiya College of Engineering, 2018.
- [6] ECEN 766 team project proposal and progress update materials, 'Brain MRI Tumor Classification with ROI Segmentation,' Texas A&M University, 2026.

Tamil Subbarayakgounder contributed dataset acquisition and preprocessing, including image resizing, normalization, filtering, and consistency checks. Yuvaraj Suresh Babu contributed ROI segmentation implementation and validation across MRI samples, including thresholding, morphological operations, and connected-component-based ROI selection. Karthigeyan Muralidharan contributed feature extraction, classifier training, integration testing, and performance analysis, including DWT/PCA feature extraction and SVM/CNN/IBK comparison. All team members contributed to report preparation, result interpretation, and final presentation material.

Appendix A. Final Gantt Chart

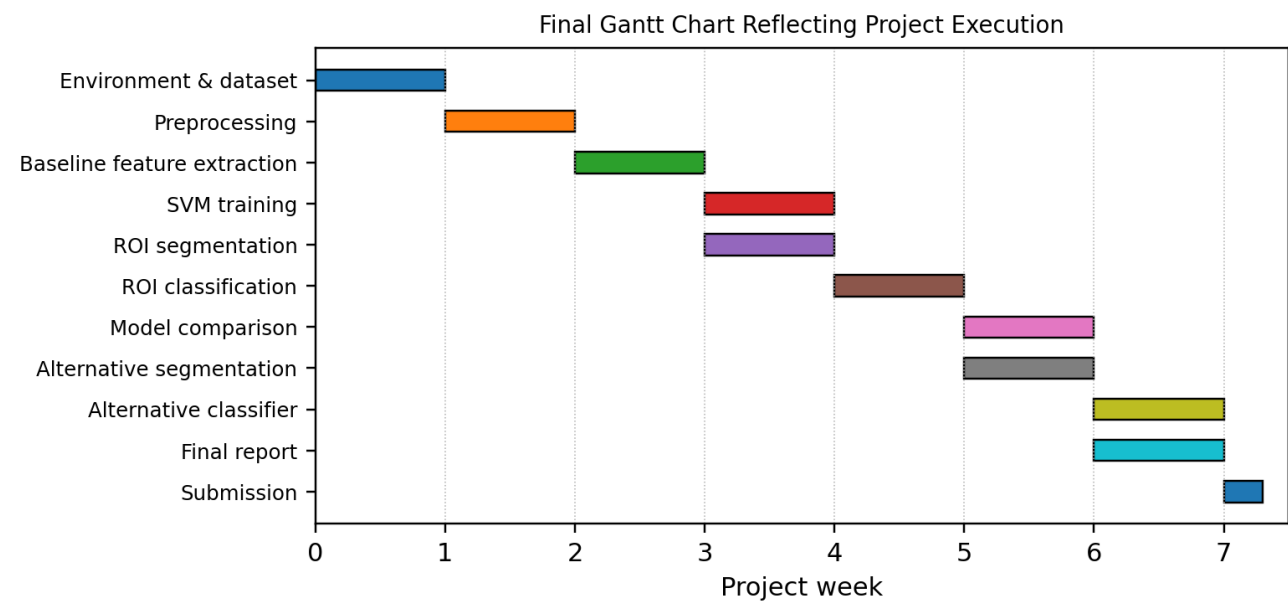


Figure A1. Final Gantt chart reflecting execution of the course project.

Appendix B. Supplemental Tables

Challenge	Limitation	Mitigation / future action
Threshold-sensitive segmentation	Otsu and watershed can fail with intensity variability	Use adaptive thresholding, PSO tuning, or U-Net segmentation
High-dimensional features	DWT/GLCM/statistical descriptors can be redundant	Apply PCA thresholds, feature importance, and ablation tests
Small or imbalanced data	Accuracy alone can hide minority-class failures	Use stratified folds and report precision, recall, and F1-score
Clinical generalization	Course dataset does not prove deployment readiness	Validate on external datasets with patient-level splits

Table B1. Limitations and mitigations summarized for future work.

Appendix C. Additional Results and Execution Screenshots

This appendix expands the evidence package for the Results section. The purpose is to place qualitative execution screenshots next to the quantitative findings in a controlled way. The screenshots are not used to claim new clinical performance; instead, they verify that the MATLAB workflow executed through the intended stages and that the reported method combinations correspond to visible segmentation, feature extraction, classification, and evaluation outputs.

C.1 MATLAB GUI workflow and execution checkpoints

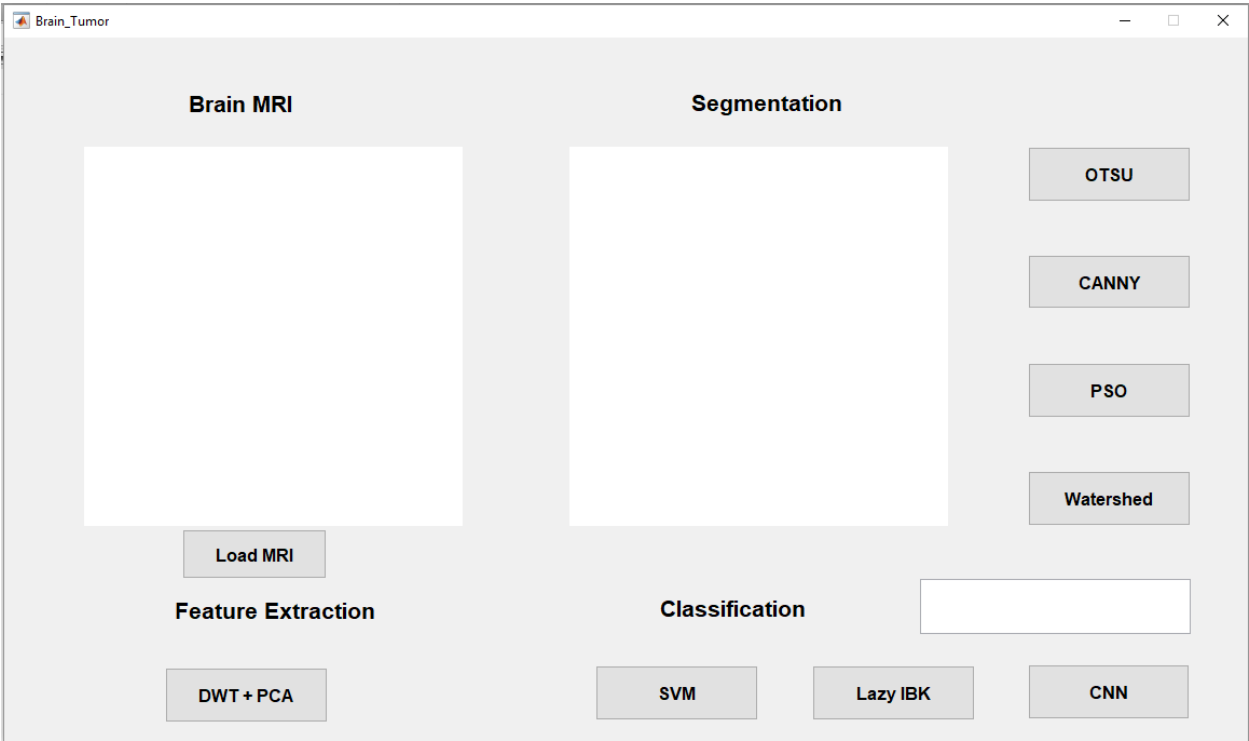


Figure C1. MATLAB GUI structure used for system-level validation. The interface exposes MRI loading, segmentation choices, DWT + PCA feature extraction, and classifier selection.

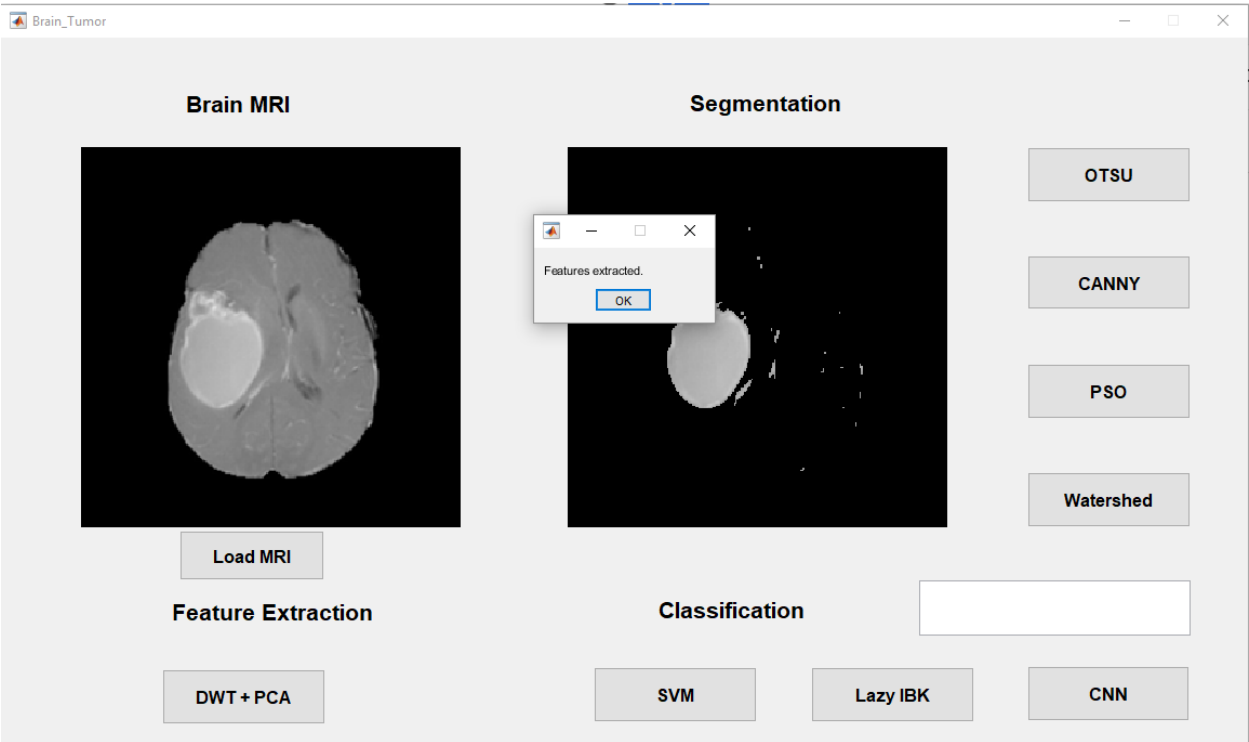


Figure C2. Feature-extraction execution checkpoint. The GUI confirms DWT + PCA feature extraction after a segmented tumor ROI is produced.

The following screenshots were placed in the Results appendix because they show the final output state of the workflow after segmentation and classification. They provide qualitative confirmation that the same GUI was used for both benign and malignant examples.

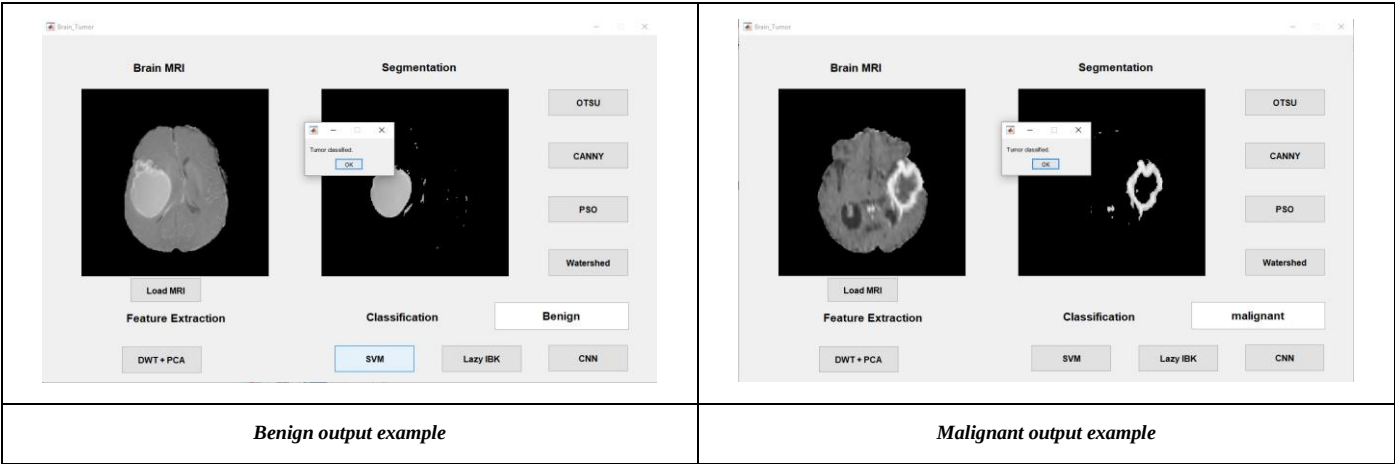


Figure C3. Representative GUI classification outputs for benign and malignant MRI examples.

C.2 Qualitative segmentation outputs

The segmentation screenshots are included to clarify why segmentation quality was treated as a bottleneck in the Results discussion. Edge-based methods may detect boundaries without fully filling the tumor region, threshold-based methods can be sensitive to intensity variation, and PSO produced visually cleaner candidate masks in the attached execution evidence.

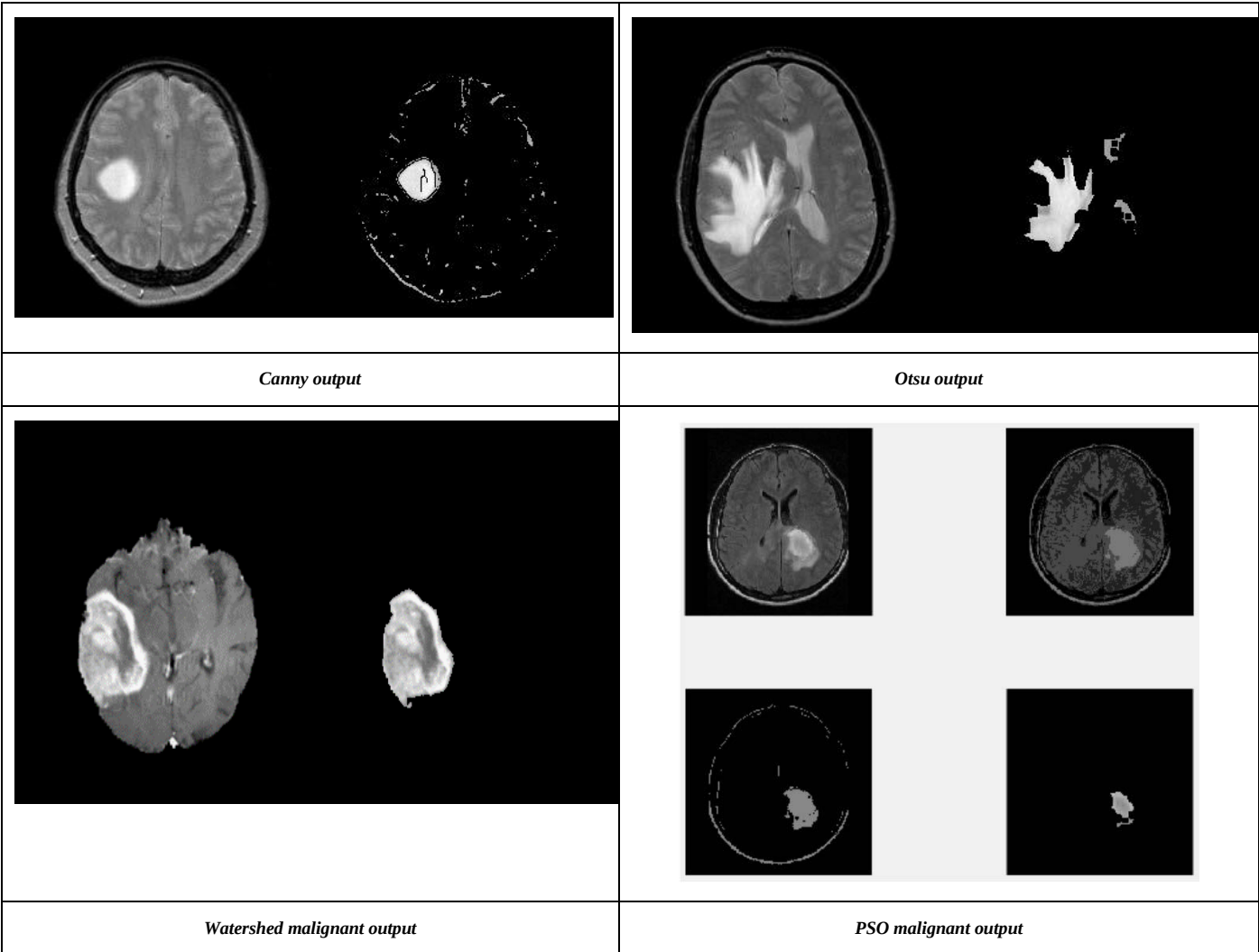


Figure C4. Representative segmentation outputs used to validate the ROI stage across different algorithms.

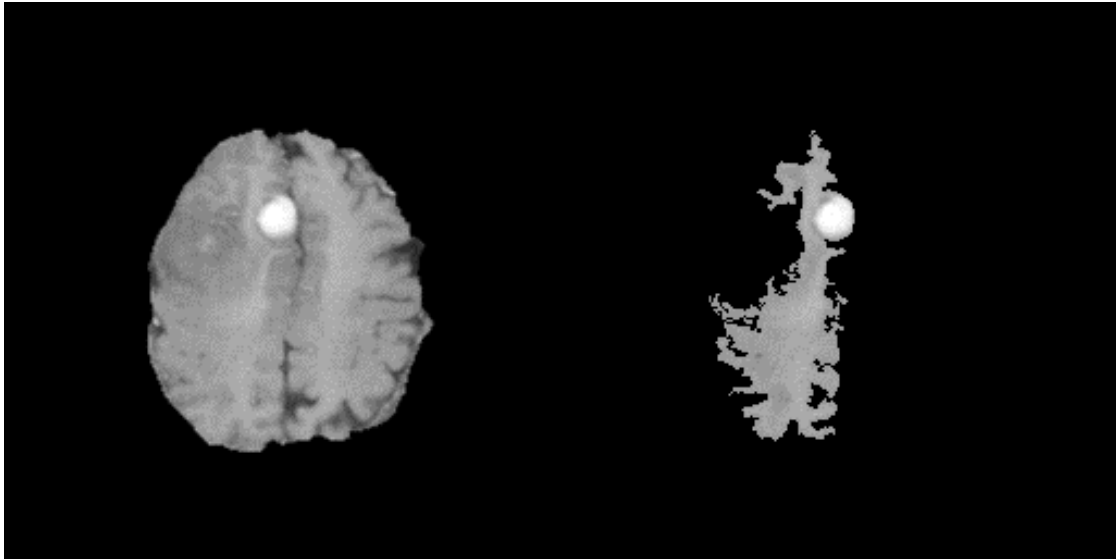


Figure C5. Canny segmentation behavior on a benign example. The screenshot illustrates why edge-only segmentation can introduce irregular masks and therefore may require cleanup before feature extraction.

C.3 CNN training-progress evidence and class-level evaluation

CNN combinations achieved the strongest reported accuracy across the tested combinations. The training-progress screenshots below are included as execution evidence for representative CNN experiments. The confusion matrices are then added as class-level result artifacts, complementing the aggregate accuracy chart.

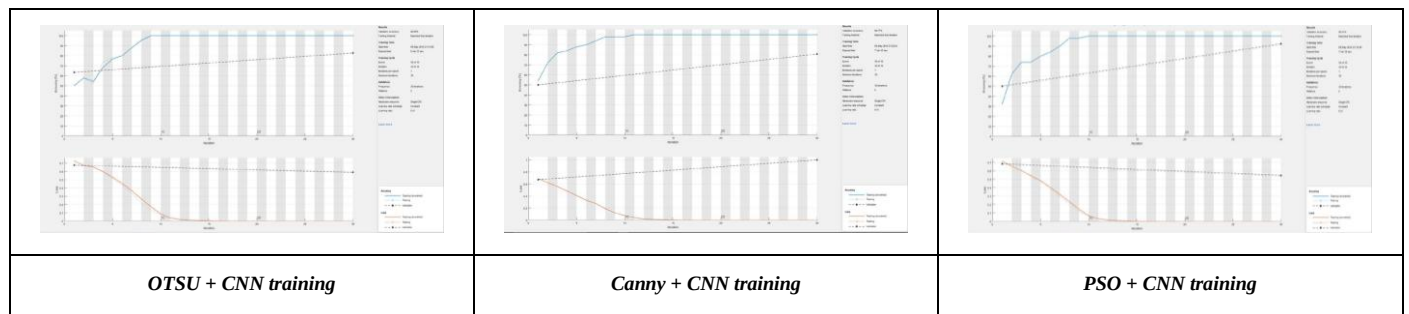
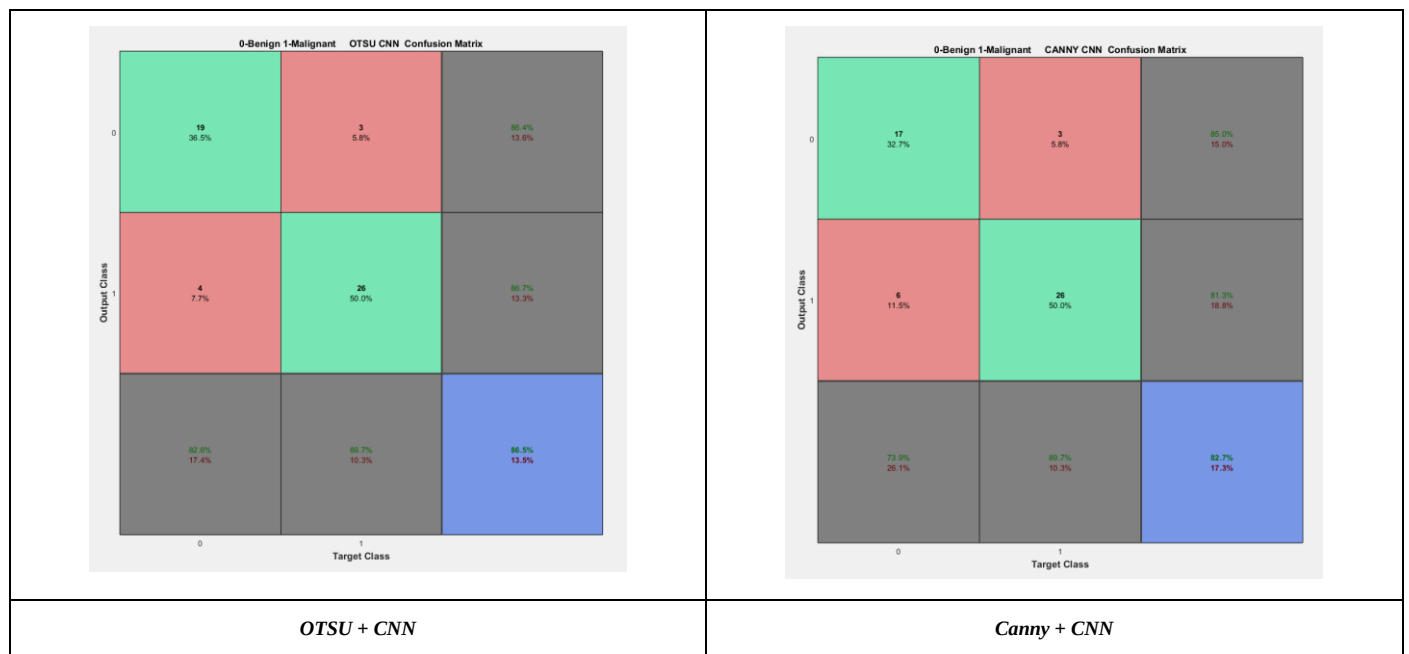


Figure C6. Training-progress screenshots for representative CNN-based experiments.



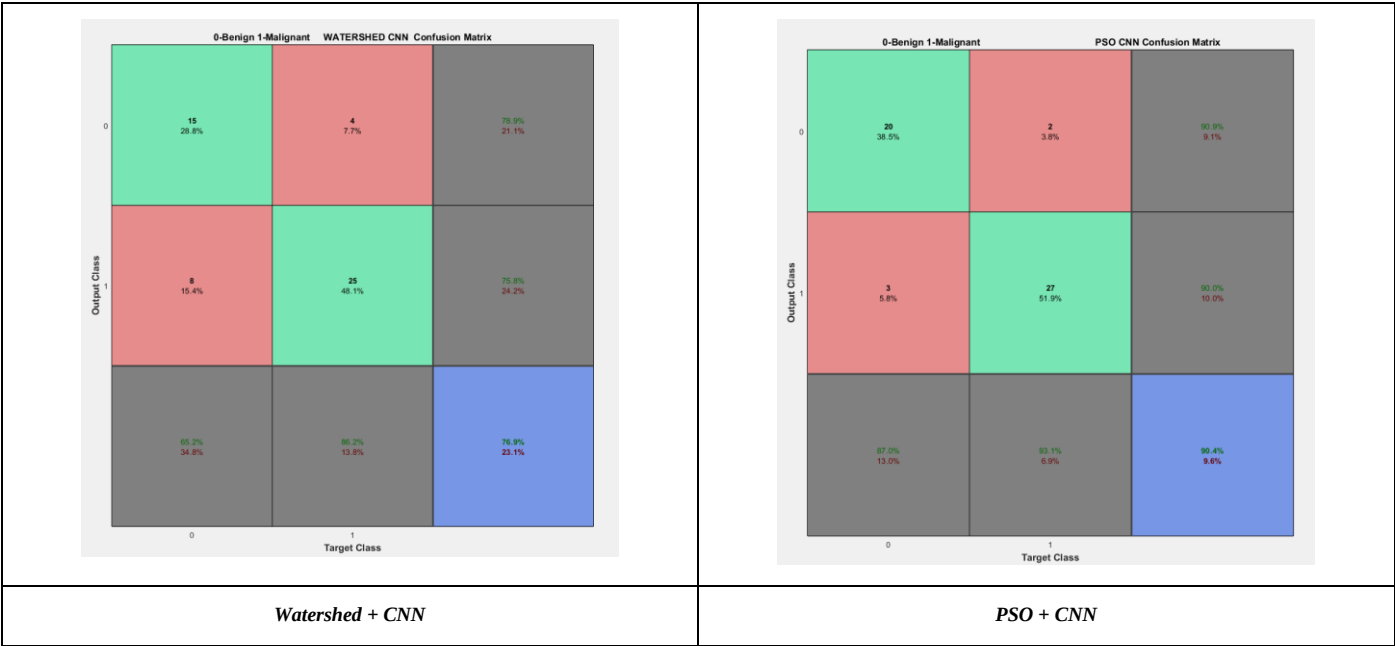


Figure C7. Representative confusion-matrix screenshots for CNN-based segmentation/classification combinations.

C.4 Expanded quantitative result placement

For clarity, the full integration-testing result table is repeated here next to the execution screenshots. The strongest reported combination is PSO + CNN at 91%, while Lazy IBK remains near 44-45% across segmentation methods, supporting the discussion that classifier capacity and segmentation quality jointly determine the final result.

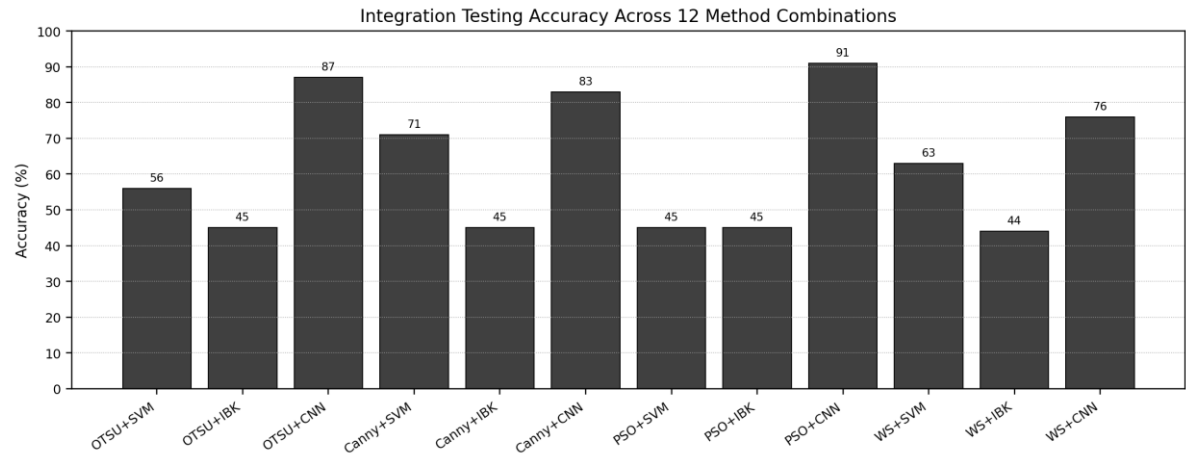


Figure C8. Accuracy comparison across all segmentation and classifier combinations. The chart supports the main Results conclusion that PSO + CNN produced the highest reported accuracy.